Chapter 13
How to Cope with Disturbances in Biped Locomotion?

Miomir Vukobratović, Branislav Borovac, Mirko Raković and Milutin Nikolić

Abstract It is expected that the humanoid robots of the near future will 'live' and work in a common environment with humans. This imposes the requirement that their operational efficiency ought to be close to that of men. The main prerequisite to achieve this is to ensure the robot's efficient motion that is its ability to compensate for the ever-present disturbances. The work considers the strategies of how to compensate for the disturbances of different intensities: small which are permanently present and large that jeopardize the robot's dynamic balance instantly. It was illustrated that those two classes of disturbances require quite different compensation approaches.

13.1 Introduction

Although the problem of bipedal gait has been in the focus of researchers for almost 40 years, not all aspects of the gait synthesis and control have been solved yet in a satisfactory way. Besides, as the moment of regular involvement of humanoid robots in everyday human activities is getting closer, the quality of their behavior (where anthropomorphism is one of important aspects) is of growing importance. It is widely accepted that the future home or personal robots will become more and more human-like regarding motion, intelligence, and communication [1]. However,

M. Vukobratović
Mihajlo Pupin Institute, Volgina 15, 11000 Belgrade, Serbia
e-mail: vuk@robot.imp.bg.ac.rs

B. Borovac (✉) · M. Raković · M. Nikolić
Faculty of Technical Sciences, University of Novi Sad, Trg Dositeja Obradovića 6, 21000 Novi Sad, Serbia
e-mail: borovac@uns.ac.rs

M. Raković
e-mail: rakovicm@uns.ac.rs

M. Nikolić
e-mail: milutinn@uns.ac.rs

© Springer International Publishing Switzerland 2016

G.M. Dimirovski (ed.), *Complex Systems*, Studies in Systems, Decision and Control 55, DOI 10.1007/978-3-319-28860-4_13

milutinn@uns.ac.rs

basic requirement is to ensure motion, but it has to be clear that it is subjected to various but permanent disturbances This paper is concerned with humanoid's "reaction" to the disturbances, i.e., how to cope with them during the gait. Of course, the compensation mode will depend on the type of disturbance. Both, gait synthesis and its realization is based on use of Zero Moment Point (ZMP) [2–7].

Basic requirements to be fulfilled in a successful realization of the gait should be the simultaneous realization of a coordinated and functional motion of the joints (realization of the given gait type) and constant preservation of dynamic balance. All robot's movements are controlled in joint state space (*internal synergy*), whereas the verification of the realization efficiency is based on the robot's behavior in the external (Cartesian) coordinates (*external synergy*). In order to have the control task realized in one and verified in another state space it is necessary to have a unique relationship between these two spaces. During the regular gait this unique relationship is ensured by fulfilling the requirement for preserving dynamic balance, which is manifested as the requirement that at least terminal link of the kinematic chain of the supporting leg (or of both legs in the case of double-support phase) is immobile with respect to the ground.

13.2 Basic Issues

It is accustomed to compare the performances of humanoid robots with those of humans. This is especially related to the gait and compensation of disturbances arising during its realization. However, it should be borne in mind that despite of the apparent simplicity and ease of its realization, the gait is a complex activity. Man, at the beginning of its life, learns how to walk during several months [8] (this stage may also be called *basic learning*), whereas afterwards he practices walk all the time, constantly improving and refining it. This is related not only to the realization of basic walking patterns (say walking forward on a flat surface, climbing, and descending the staircases, etc.), but also to learning and perfecting the reactions to all kinds of disturbances that inevitably arise during the gait.

The main task of a locomotion system is to maintain the gait, i.e., to "keep on the feet," either standing or walking. For a system whose gait is not jeopardized we say that it is dynamically balanced. In the presence of disturbances (it should be emphasized that small disturbances are always present and cannot be avoided) the man reacts so as to remove the effect of the disturbance in the way that affects the least the realization of the activity that has been pursued before the occurrence of the disturbance. Of course, if the disturbance is of too high an intensity and man (humanoid) is under the threat of immediate fall, the whole attention has to be paid to its prevention and the realization of the former activity is to be abandoned.

The main indicator of dynamic balance of a bipedal robot is the ZMP. All the time the ZMP is within the support area, excluding the edges. (both in the single–and double-support phase), the system will be dynamically balanced. The action of the disturbance causes displacement of the ZMP from its reference

milutinn@uns.ac.rs

position to the edge of the supporting foot, and the action of the compensating movements ought to be in the opposite direction, to return it close to the reference position (i.e., to bring it to the "safety zone").

Disturbances may differ in respect of both the type and action on the humanoid robot. As far as we know, their only categorization [9] which will be briefly repeated here, distinguishes small, medium and large disturbances. The categorization is based on the effect by which the disturbance jeopardizes dynamic balance and possibility of the continuation of the realization of the motion performed to the instant of the disturbance occurrence. In all three cases, it is assumed that the humanoid in the beginning of the considered period performs the reference motion, and then the disturbance begins to act.

(i) Because of the action of external disturbance force of **smaller intensity**, ideal tracking of joint trajectories is disturbed, as well as the ZMP position, but in such a manner that it still remains within a "safety zone." Appropriate control actions can return the system to the reference trajectory.
(ii) Assume now that a disturbance force of **medium intensity** is involved. In this case too, the force disturbs the ideal tracking of joint trajectories, and increases the deviation of the ZMP from its reference position. To preserve dynamic balance, the humanoid must undertake a 'more resolute' action (e.g., arms swinging), in order to ensure that the ZMP remains within the 'safety zone' and the system returns to the reference motion.
(iii) In the case of a **high-intensity** disturbance, as mentioned earlier, unpowered (passive) DOFs arise, and the system as a whole starts to rotate about the edge of the support area. The attempt to minimize deviations of joint trajectories and resume reference motion is senseless if the system has lost the dynamic balance. Hence, it is of highest interest to preserve (or reestablish) dynamic balance, taking no care at the moment of joints trajectories tracking. Because of that, the humanoid has to abandon the realization of the previous reference motion and, for example, step by one leg in the direction of falling, support on it, and—in the next several steps—return to the reference trajectory.

In all cases, control task is to minimize the deviations of the real humanoid state from the reference one for all the joints, while preserving dynamic balance whereby, especially in third case, the priority is to prevent the system from falling down, i.e., to preserve its dynamic balance [10].

In practice, small disturbances are most common, so that the problem of gait control reduces to case 1, which has been investigated in details in [11].

In the case of the compensation of large disturbances it is necessary to take care of two requirements that are to be simultaneously satisfied by the compensating movement:

(i) Forces induced by the compensating movement have to act so that the ZMP position, usurped by the action of the disturbance, is maintained within the support area.

milutinn@uns.ac.rs

(ii) Bearing in mind that the action of the forces induced by the compensating movement lasts short (it ends immediately after the ending of the compensating movement) *the humanoid's position at the end of the compensating movement should be such to ensure dynamic balance without the action of the forces induced by the motion*. This means that the projection of the system's CM at the end of the compensating movement has to be within the support area.

These two requirements impose serious constraints. Namely, compensating movements are often very complex (a number of joints move synchronously) and man learns them in the course of mastering the skill of walking by the trial-and-error procedure and then selects and adopts them. Hence, it is plausible to observe the characteristic ways of disturbance compensation used by the man, and then, by analyzing them, "discover" the reasons why man uses them as such.

13.3 Biped Mechanical Structure

The structure of the basic mechanism having 36 DOFs and one-link trunk, used in the present work, is shown in Fig. 13.1. The first kinematic chain represents the legs (links 1–27), the second chain extends from the pelvis and comprises the trunk and the right hand (links 28–33), and the third chain (links 34–36) forms the left shoulder and arm.

The multi-DOF joints were modeled as a set of "fictitious" links (massless links of zero-length) interconnected with the joints having one DOF. For example, the hip joints, which are in reality spherical joints with three DOFs, are modeled as sets of three one-DOF joints whose axes are mutually orthogonal. Thus, the right hip is modeled by a set of simple joints 13, 14, and 15 (with the unit vectors of rotation axes $\boldsymbol{e_{13}}$, $\boldsymbol{e_{14}}$, and $\boldsymbol{e_{15}}$), and the left hip by the set of joints 16, 17, and 18 (the unit vectors $\boldsymbol{e_{16}}$, $\boldsymbol{e_{17}}$, and $\boldsymbol{e_{18}}$). The links connecting these joints (for the right hip the links 13 and 14, and for the left links 16 and 17) were needed only to satisfy the mathematical formalism of modeling a kinematic chain. The other links (those that are not part of the joints with more DOF's) whose characteristics correspond to the links of an average human body (link 9 corresponds to the shank, link 12 to the thigh, link 30 to the trunk, etc.), are presented by solid lines in Fig. 13.1. In the same figure, the links that were needed only for modeling "complex" joints with more DOFs (having no mass and with the moment of inertia and length being equal zero) are presented by dashed lines, to indicate their "fictitious" nature.

Of special importance is the way of modeling the foot-ground contact in order to determine the exact position of the ZMP during the motion and observe the moment when the mechanism is out of dynamic balance. The loss of dynamic balance means that the mechanism collapses by rotating about one of the edges of the supporting foot, and this situation, obviously, has to be prevented. The contact of the mechanism with the ground is modeled by two rotational joints, determined with the unit vectors $\boldsymbol{e_1}$ and $\boldsymbol{e_2}$ (Fig. 13.1), mutually perpendicular. At the ZMP for dynamically balanced motion, it is constantly ensured that $\mathbf{M_Y} = 0$ $((\mathbf{M_X} \perp \mathbf{M_Y}) \wedge (\mathbf{M_X}, \mathbf{M_Y} \in$

milutinn@uns.ac.rs

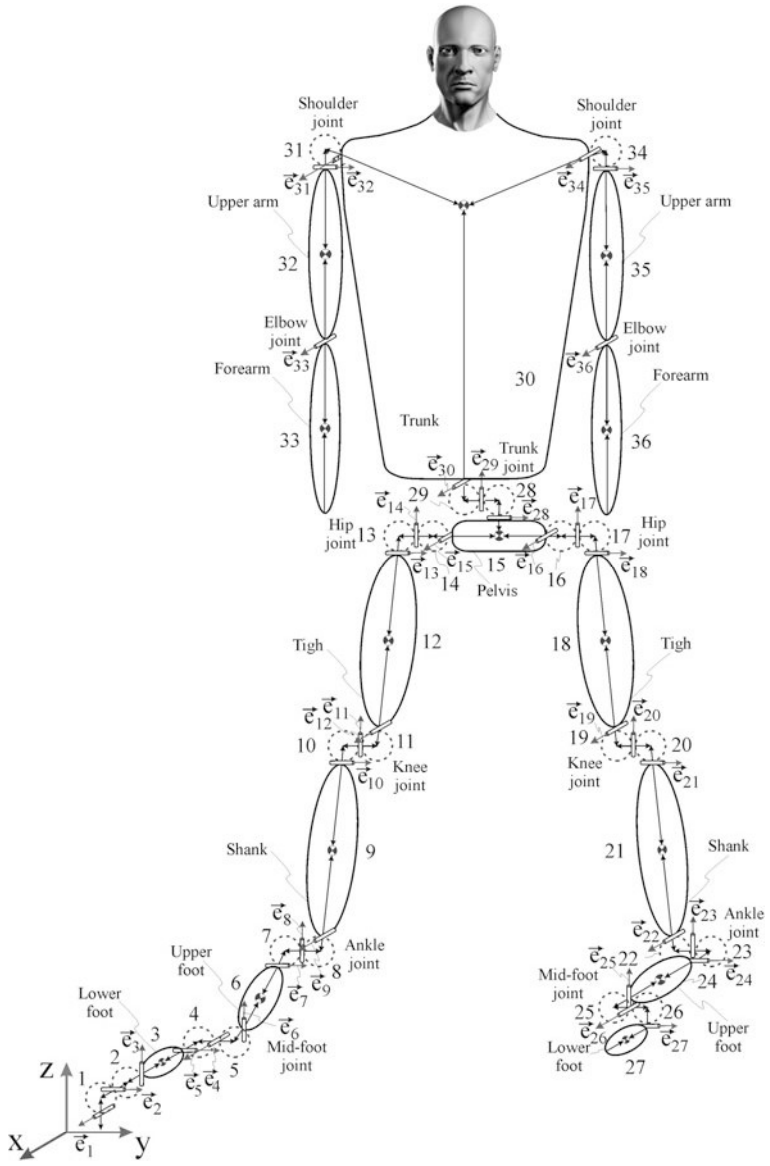


Fig. 13.1 Schematic of the robot's mechanical configuration

XoY)). It should be especially emphasized that the mechanism feet were modeled as the two-link ones. In Fig. 13.1 the anterior part of the right foot (toes) is presented by link 3, and its main part (foot body) by link 6. The toes of the left foot are presented by link 27 and the foot body by link 24. The trunk is customarily considered as one-link rigid body. However, since the trunk's base is the flexible

milutinn@uns.ac.rs

spinal column, the assumption of one-link rigid trunk is not quite justified. To investigate the influence of the trunk bending on the synthesized motion of the mechanism and its role in control, the trunk was modeled as being multilink. The trunk was divided into several links (in this work it was modeled as being composed of 10 links) interconnected with the joints having two DOFs each (rotation about the y-axis (inclining forward-backward) and rotation about the x-axis (inclining left-right)). By this, the number of DOFs of the overall model was significantly increased, amounting to 54. The way of modeling the 10-link trunk and its "fitting into" the overall model of the humanoid is shown in Fig. 13.2.

In case of large disturbances system motion was simulated on our new software [12]. The pelvis link was chosen as the base one, Fig. 13.3, and its position and orientation in the space are presented by three translatory and angular coordinates $\mathbf{X} = [x, y, z, \theta, \varphi, \psi]^T$. To the pelvis are connected the other kinematic chains, of which the first chain (links 1–9) represents the right leg; the second chain (links 1, 10–17) stands for the left leg; the third chain (links 1, 18–42) represents the backbone and the right arm, whereas the fourth kinematic chain represents the 10-link trunk (the links are interconnected by 2-DOF joints) and the left arm (links

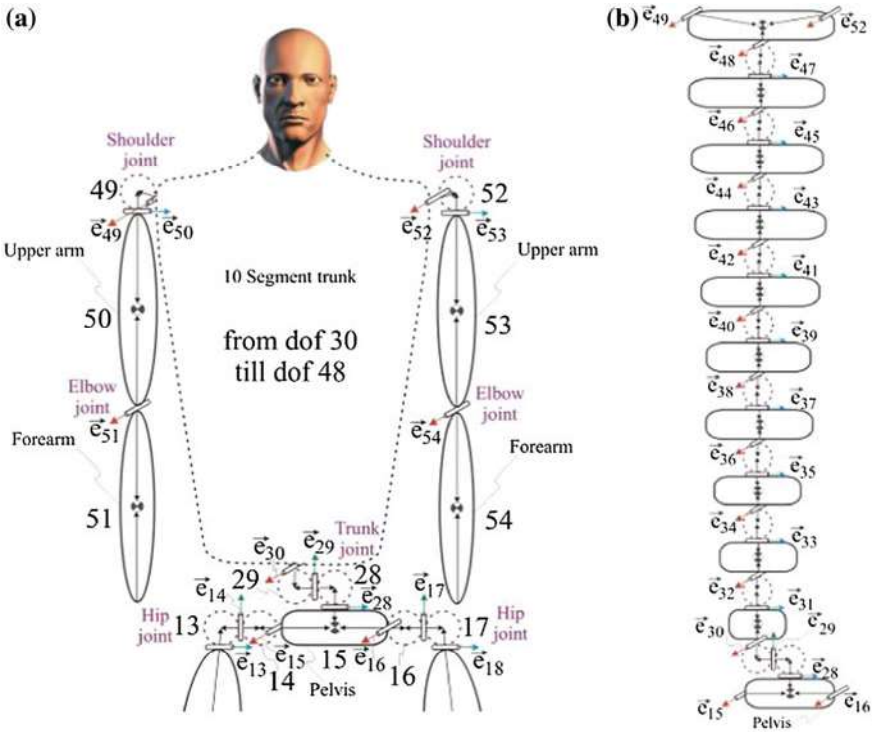


Fig. 13.2 Multi-link trunk: **a** Its "fitting into" the humanoid's model instead of one-link trunk; **b** modeling the multi-link trunk

milutinn@uns.ac.rs

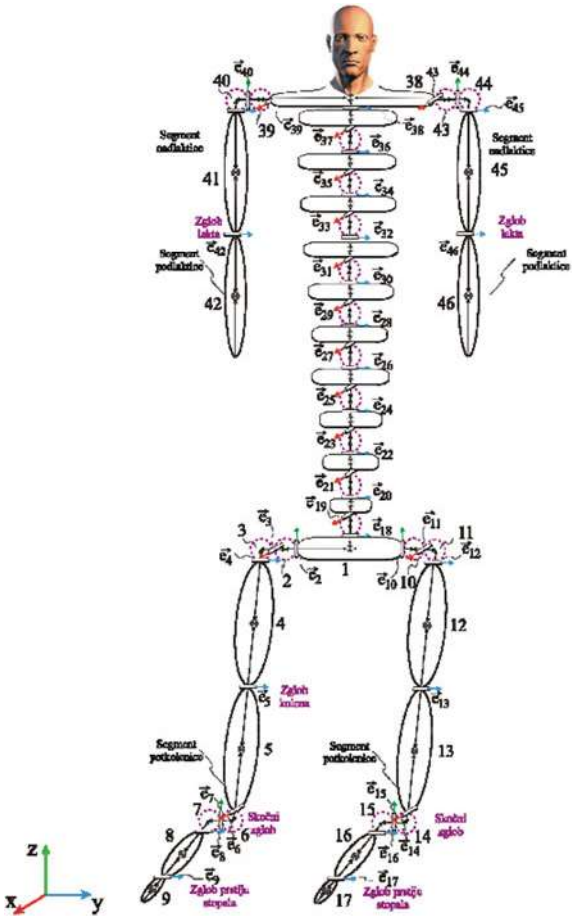


Fig. 13.3 Mechanical structure of the robot with 46 links and 45 joints

1, 18–38, 43–46). The head is included into the link 38. The feet are two-link ones, whereby between the foot body and toes there exists only one DOF (left foot: links 8 and 9; right foot: links 16 and 17).

As already described, the motion the ZMP trajectory is prescribed. Different ZMP trajectories may result in completely different motions of the trunk. In this work, the ZMP path at the foot-ground contact surface divided into six segments was adopted in advance. Then, the ZMP traveling time along each particular segment was varied in such a way that the overall step duration remained unchanged. For each of these cases, the trunk motion was synthesized. The motion synthesis was performed for the mechanism with a one-link trunk and two-link feet (the mechanism with 36 DOFs), shown in Fig. 13.1. The basic ZMP path (which includes both the single- and double-support phase) was adopted as shown in Figs. 13.4 and 13.5. Time dependence of the ZMP position during the half-step was realized in the following way. The path was divided into six segments, in each of

milutinn@uns.ac.rs

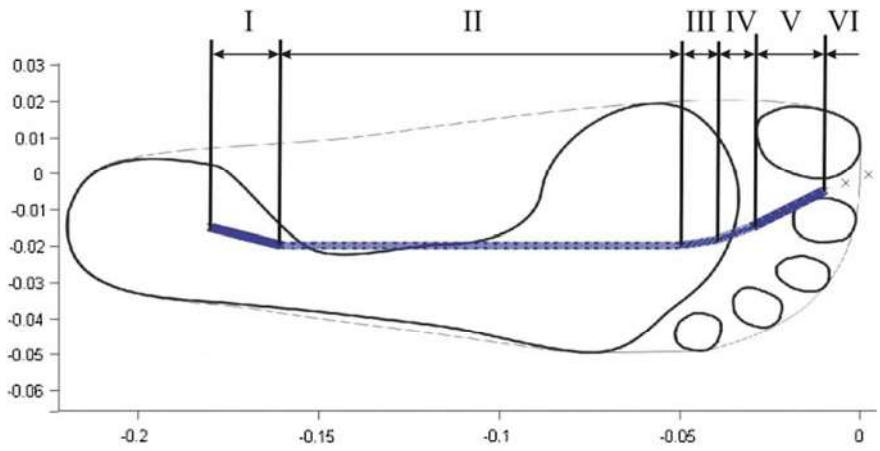


Fig. 13.4 Reference ZMP path under foot

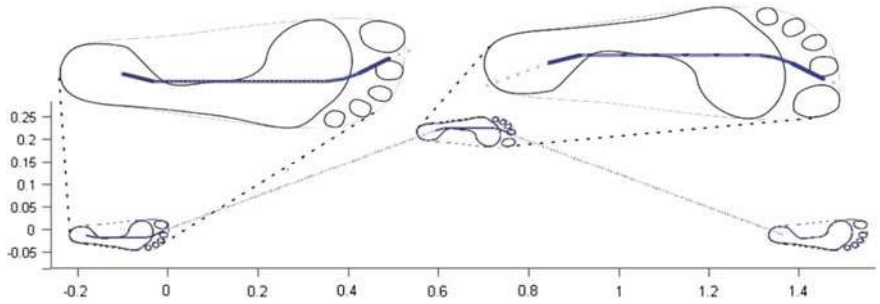


Fig. 13.5 Reference ZMP path for full step

them the ZMP, traveling from the starting to the final position, remained for a certain time. In segment I ZMP remained 10 % of the overall step duration, in II 42 %, in III 4 %, in IV 4 %, in V 19 % and in VI 21 %.

13.4 Compensation of Small Disturbances

Any deviation of internal synergy from the reference caused by the disturbance action, produces also deviation of the ZMP position from its reference, which can endanger dynamic balance. In this work, we consider disturbance in the form of the force action at the humanoid's right shoulder (Fig. 13.6). Figure 13.6a shows a perspective view of the action of the force (components F_x and F_y), Fig. 13.6b presents the change of the intensity of the force (components F_x and F_y) during a half-step. The half-step lasts 0.8 s, the force starts to act 0.15 s after the beginning of

milutinn@uns.ac.rs

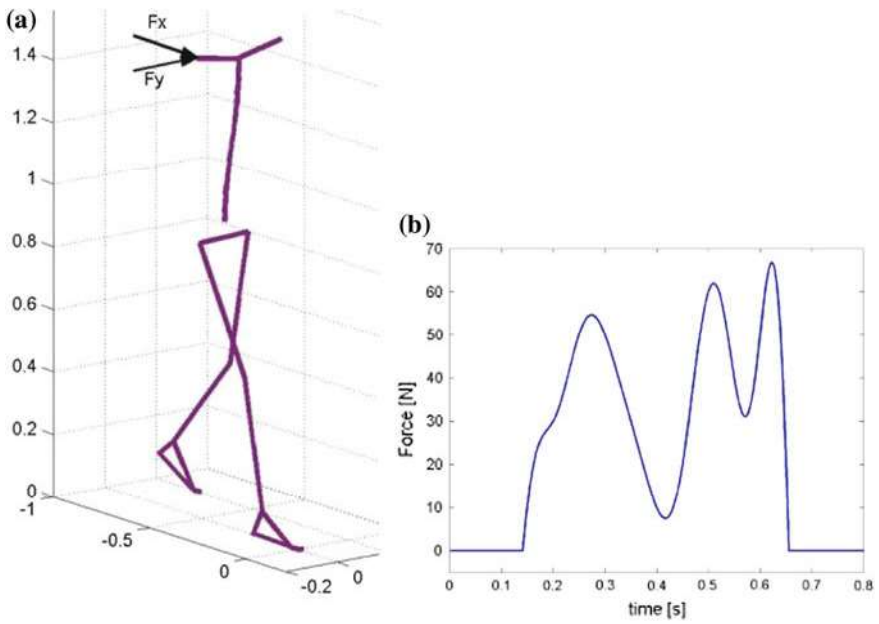


Fig. 13.6 **a** Force action at the mechanism's shoulder, **b** time profile of the force intensity

motion, and its action terminates after 0.65 s. During the second half-step the force does not act. However, in view of the disturbance of internal synergy in the beginning of the second half-step, there exists the disturbance of initial conditions type. The control system attempts to bring the internal synergy closer to the reference and preserve simultaneously dynamic balance. We consider the cases of acting of only component F_x, only component F_y, and both components simultaneously.

Generally speaking, in each time instant t_i, the total control signal at each of the joints consists of the reference control (calculated for the case of motion without disturbances) and corrections, which depend of the disturbance intensity. In other words:

$$u_{n_i}^{(t_i)} = u_{\text{reference } n_i}^{(t_i)} + \Delta u_{\text{at joint } n_i}^{(t_i)}.$$

The corrective part $\Delta u_{\text{at joint } n_i}^{(t_i)}$ consists also of two parts:

- One part of the corrective control serves to preserve dynamic balance, i.e., to minimize the ZMP position deviation from the reference $\Delta u_{\text{ZMP at joint } n_i}^{(t_i)}$ one. This task can be allocated to one or more mechanism's joints.
- The other part of the corrective control ($\Delta u_{\text{local at joint } n_i}^{(t_i)}$) should ensure minimization of the deviation of the actual synergy from the reference one at each joint. This part of the corrective control we call local control, as the regulators involved act locally, i.e., only at the individual joints.

milutinn@uns.ac.rs

Hence, the overall compensation (at each of the joints) can be written in the following way:

$$\Delta u_{n_i}^{(t_i)} = \Delta u_{\text{ZMP at joint } n_i}^{(t_i)} + \Delta u_{\text{local at joint } n_i}^{(t_i)}. \tag{13.1}$$

In the above expressions n_i refers to the ith joint of the mechanism configuration having in all n joints. Naturally, each joint tends to preserve its motion as close as possible to the desired one, so that the task of forming local corrective control may be imposed on all the joints where there exist deviations from the desired motion. The task of compensating for ZMP deviation can not be (and is not necessarily) realized at every joint at the same time, but the task may be allocated to only some joints. In this, it should be borne in mind that the axis of the joint performing compensation is perpendicular to the direction of ZMP deviation. In other words, the ZMP deviation in the x-direction may be compensated for only by the joints whose axes are parallel to the y-direction, and the ZMP deviations in the y-direction by the joints whose axes are parallel to the x-direction. Therefore, if we want to compensate for the ZMP deviations in the x-direction with the aid of the joint n_i, the control law will be of the form:

$$\Delta u_{\text{ZMP at joint } n_i}^{(t_i)} = k_{p\,\text{ZMP}\,n_i} \cdot \Delta \text{ZMP}_x^{(t_i)} + k_{i\,\text{ZMP}\,n_i} \cdot \sum_{i=1}^{t_i} \Delta \text{ZMP}_x^{(i)} + k_{d\,\text{ZMP}\,n_i} \cdot \left(\Delta \text{ZMP}_x^{(t_i)} - \Delta \text{ZMP}_x^{(t_i-1)} \right), \tag{13.2}$$

where $k_{p\ \text{ZMP}\ ni}$ represents the position feedback gain at the joint n_i, for the compensation of ZMP deviation; $k_{i\ \text{ZMP}\ ni}$ and $k_{d\ \text{ZMP}\ ni}$ represent integral and derivative feedback gains at the joint n_i, also for compensating the ZMP deviation; $\Delta \text{ZMP}_x^{(t_i)}$ stands for the deviation of ZMP in the direction of the x-axis at a time instant t_i.

The law of local control to compensate for the deviations from reference values at the joint n_i can be written as:

$$\Delta u_{\text{local } i \text{ at joint } n_i}^{(t_i)} = k_{p\,n_i\,\text{local}} \cdot \Delta q_{n_i}^{(t_i)} + k_{i\,n_i\,\text{local}} \cdot \sum_{i=1}^{t_i} \Delta q_{n_i}^{(i)} + k_{d\,n_i\,\text{local}} \cdot \left(\Delta q_{n_i}^{(t_i)} - \Delta q_{n_i}^{(t_i-1)} \right). \tag{13.3}$$

Analogously to the previous case, $k_{p\,n_i\,\text{local}}$ represents the position feedback gain at the joint n_i for the compensation of its deviation from the reference value, whereas $k_{i\,n_i\,\text{local}}$ and $k_{d\,n_i\,\text{local}}$ represent the integral and derivative feedback gains, also for the compensation of deviations at the joint n_i from the reference values.

Let us consider now in more detail how ZMP deviations are compensated for. The ZMP deviation in the x-direction may be compensated for by the ankle (joint 7), hip (joint 13), or the waist (joint 28) (see Fig. 13.1). Equation (13.2), applied to any of these cases, would be of the form:

milutinn@uns.ac.rs

$$\Delta u^{(t_i)}_{\text{ZMP at joint 7}} = k_{p\,\text{ZMP 7}} \cdot \Delta \text{ZMP}_x^{(t_i)} + k_{i\,\text{ZMP 7}} \cdot \sum_{i=1}^{t_i} \Delta \text{ZMP}_x^{(i)} + k_{d\,\text{ZMP 7}} \cdot \left(\Delta \text{ZMP}_x^{(t_i)} - \Delta \text{ZMP}_x^{(t_i-1)}\right), \tag{13.4}$$

$$\Delta u^{(t_i)}_{\text{ZMP at joint 13}} = k_{p\,\text{ZMP 13}} \cdot \Delta \text{ZMP}_x^{(t_i)} + k_{i\,\text{ZMP 13}} \cdot \sum_{i=1}^{t_i} \Delta \text{ZMP}_x^{(i)} + k_{d\,\text{ZMP 13}} \cdot \left(\Delta \text{ZMP}_x^{(t_i)} - \Delta \text{ZMP}_x^{(t_i-1)}\right), \tag{13.5}$$

$$\Delta u^{(t_i)}_{\text{ZMP at joint 28}} = k_{p\,\text{ZMP 28}} \cdot \Delta \text{ZMP}_x^{(t_i)} + k_{i\,\text{ZMP 28}} \cdot \sum_{i=1}^{t_i} \Delta \text{ZMP}_x^{(i)} + k_{d\,\text{ZMP 28}} \cdot \left(\Delta \text{ZMP}_x^{(t_i)} - \Delta \text{ZMP}_x^{(t_i-1)}\right). \tag{13.6}$$

In the case, we want to compensate for the ZMP deviation in the y-direction, the expressions to calculate compensational control become:

$$\Delta u^{(t_i)}_{\text{ZMP at joint 9}} = k_{p\,\text{ZMP 9}} \cdot \Delta \text{ZMP}_y^{(t_i)} + k_{i\,\text{ZMP 9}} \cdot \sum_{i=1}^{t_i} \Delta \text{ZMP}_y^{(i)} + k_{d\,\text{ZMP 9}} \cdot \left(\Delta \text{ZMP}_y^{(t_i)} - \Delta \text{ZMP}_y^{(t_i-1)}\right), \tag{13.7}$$

$$\Delta u^{(t_i)}_{\text{ZMP at joint 15}} = k_{p\,\text{ZMP 15}} \cdot \Delta \text{ZMP}_y^{(t_i)} + k_{i\,\text{ZMP 15}} \cdot \sum_{i=1}^{t_i} \Delta \text{ZMP}_y^{(i)} + k_{d\,\text{ZMP 15}} \cdot \left(\Delta \text{ZMP}_y^{(t_i)} - \Delta \text{ZMP}_y^{(t_i-1)}\right), \tag{13.8}$$

$$\Delta u^{(t_i)}_{\text{ZMP at joint 30}} = k_{p\,\text{ZMP 30}} \cdot \Delta \text{ZMP}_y^{(t_i)} + k_{i\,\text{ZMP 30}} \cdot \sum_{i=1}^{t_i} \Delta \text{ZMP}_y^{(i)} + k_{d\,\text{ZMP 30}} \cdot \left(\Delta \text{ZMP}_y^{(t_i)} - \Delta \text{ZMP}_y^{(t_i-1)}\right). \tag{13.9}$$

The local control to minimize deviations at each particular joint from the reference can be written in the following way:

$$\Delta u^{(t_i)}_{\text{local at joint}n_i} = k_{p\,n_i\,\text{local}} \cdot \Delta q_{n_i}^{(t_i)} + k_{i\,n_i\,\text{local}} \cdot \sum_{i=1}^{t_i} \Delta q_{n_i}^{(i)} + k_{d\,n_i\,\text{local}} \cdot \left(\Delta q_{n_i}^{(t_i)} - \Delta q_{n_i}^{(t_i-1)}\right). \tag{13.10}$$

The feedback gains of local regulators and for preserving dynamic balance, k_p, k_i, and k_d, are not constant but their values vary in a certain range in dependence of the instantaneous position of the ZMP. The values for k_p, k_i, and k_d are determined in a heuristic manner. The change in the values of the feedback gains must not be of "step function character" but a gradual one, taking place in the following way. Around the

ZMP trajectory, at a distance of 5 mm, a zone is formed. If the ZMP is within that zone, the feedback gains of the regulator for maintaining dynamic balance begin to decrease (up to the minimal values), and the coefficients of local regulators start to rise (up to the maximal values). After that they retain their (minimal and

milutinn@uns.ac.rs

maximal) values till the next change. If the ZMP is out of the 5-mm zone, the feedback gains of the regulator for maintaining dynamic balance begin to increase (up to the maximal values), and the feedback gains of local regulators start to decrease (up to the minimal values). After that they retain their (minimal and maximal) values till the next change, which will happen when ZMP crosses over 5-mm border. Values of the increments and decrements for all the feedback gains k_p, k_i, and k_d were also determined based on heuristic.

Minimal and maximal values of the feedback gains in the case of compensating for ZMP deviation and the values of increments and decrements are presented in Table 13.1, whereas Table 13.2 gives the minimal and maximal values of feedback gains, along with the values of increments and decrements for the local regulators.

In Fig. 13.7 is illustrated the simulation of humanoid's motion when the disturbance force F_x acted at the shoulder (pushing the humanoid forward) and compensation was performed by the ankle joint. The trunk was a 10-link one. Reference control was applied at all joints. Additional control for correcting ZMP deviation, acting at the joints 7 and 9 was defined by Eqs. (13.4) and (13.7), while the local regulators were defined by Eq. (13.10). In view of the fact that the disturbance force was pushing the mechanism forward and the compensation was performed at the ankle of the supporting leg, the entire system inclined forward, so that the leg being in the swing phase might kick against the ground.

In order to eliminate the inclination of the overall system forward and allow regular pace, additional compensation was performed at the hip of the leg in the swing phase (joint 18) so that the same Δu that has been added to the ankle (joint 7) of the supporting leg was added to the hip joint of the swing leg in order to increase humanoid's step in the gait direction. Such correction of the internal synergy has been called preventive correction. Although essential stepping out was made in this half-step, it is evident from the figure that the ZMP trajectory deviated only a little from the reference one. The increased step length during the first half-step was fully

Table 13.1 Feedback gains for ZMP compensation and the corresponding increments and decrements

Gain	Joint	Min.	Max.	Inc.	Dec.
$k_{p\ \mathrm{ZMP}}$	4, 7, 10, 13	1	9	0.1	0.01
$k_{p\ \mathrm{ZMP}}$	9, 15	5	13	0.1	0.01
$k_{p\ \mathrm{ZMP}}$	28–48	2	10	0.1	0.01
$k_{i\ \mathrm{ZMP}}$	All	3	3.08	0.0001	0.00001
$k_{d\ \mathrm{ZMP}}$	All	2	2.8	0.001	0.0001

Table 13.2 Feedback gains for local regulators and the corresponding increments and decrements

Gain	Joint	Min.	Max.	Inc.	Dec.
$k_{p\ \mathrm{local}}$	All	10	200	0.1	10
$k_{i\ \mathrm{local}}$	All	2	3	0.0005	0.05
$k_{d\ \mathrm{local}}$	All	3	8	0.002	0.2

milutinn@uns.ac.rs

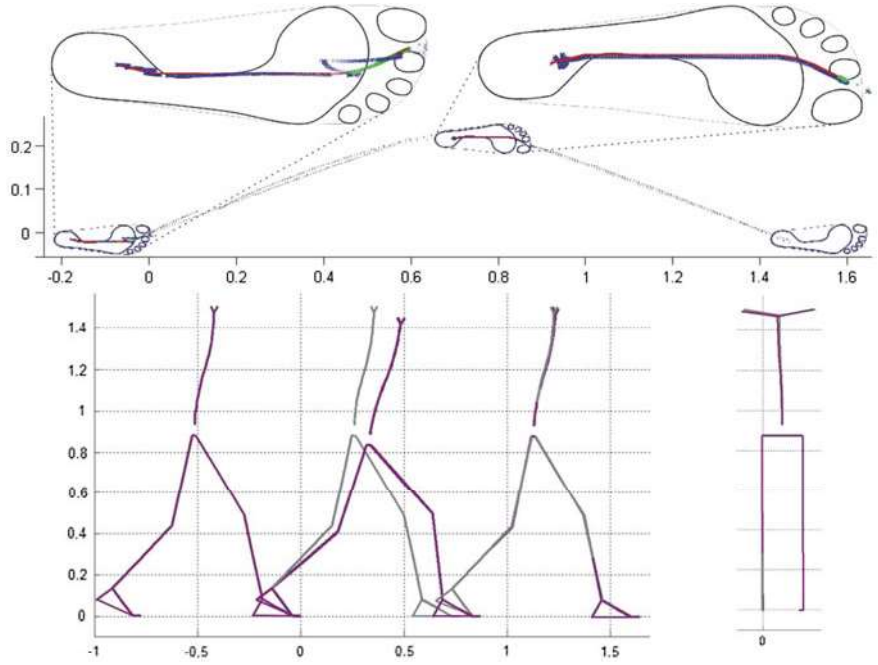


Fig. 13.7 Action of the force F_x on the mechanism shoulder. Compensation was performed by the ankle joint of the supporting leg. The *upper* picture shows the ZMP trajectory, the *lower left* a lateral view of the stick diagram, the *lower right* stick diagram in the sagittal plane at the end of the second half-step

cancelled out in the second half-step, and the rest of the humanoid mechanism too, came close to its reference position.

13.5 Compensation of Large Disturbances

The main task of any locomotion system is to "keep on the feet" (i.e., to preserve dynamic balance), either standing or walking. If the disturbance is of too high an intensity and humanoid is under the threat of immediate fall, the whole attention has to be paid to its prevention and the realization of the former activity is to be abandoned. In such a case two requirements have to be simultaneously satisfied by the compensating movement:

(i) Forces induced by the compensating movement have to act so that the ZMP position, usurped by the action of the disturbance, is maintained within the support area.

milutinn@uns.ac.rs

(ii) Bearing in mind that the action of the forces induced by the compensating movement lasts short (it ends immediately after the ending of the compensating movement) *the humanoid's position at the end of the compensating movement should be such to ensure dynamic balance without the action of the forces induced by the motion*. This means that the projection of the system's CM at the end of the compensating movement has to be within the support area.

These two requirements impose serious constraints. Namely, compensating movements are often very complex (a number of joints move synchronously) and man learns them in the course of mastering the skill of walking by the trial-and-error procedure and then selects and adopts them. Hence, it is plausible to observe the characteristic ways of disturbance compensation used by the man, and then, "discover" the reasons why man uses them as such [10].

13.5.1 Characteristic Compensating Movements

Let us consider, the case when the humanoid is standing in an upright position (Fig. 13.8a), with both feet on the ground, and on which a disturbance in the form of the impulse force is acting horizontally on its back at a height $h = 1.6$ m, whose profile is given in Fig. 13.8b. Depending on the direction and intensity of the disturbance force, there are two characteristic ways by which the humanoid may react. One is to attempt to maintain dynamic balance while not moving the feet from the ground and the other is to step in the direction of the force action. In the former case, there are two possible scenarios. The first is to generate compensating movements simultaneously at both ankles, while the whole system behaves as a unique link (Fig. 13.8c), i.e., as a single inverted pendulum. The other way (Fig. 13.8d) is to generate the compensating movements simultaneously at the ankle and the hip, but such that they act in the opposite directions. Figure 13.8d shows the

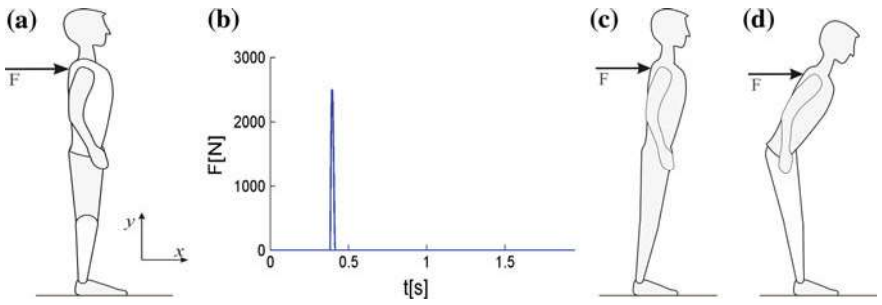


Fig. 13.8 **a** Initial posture of the humanoid **b** Profile of the disturbance force **c** Compensation by the moment at the ankle **d** Simultaneous compensation by the ankle and the hip

milutinn@uns.ac.rs

links of the legs (the lower leg and the thigh) and of the trunk (with the neck and head) as an unique link, so that the system can be approximated as a two-link one.

13.5.2 Analysis of Compensating Movements

Now, we will analyze two essentially different types of compensating movements: the case when the entire body behaves as a unique link (Fig. 13.8c) and as a two-link mechanism (Fig. 13.8d). In order to analyze both types of movements and establish their main characteristics we will form a planar model of the humanoid (Fig. 13.9) in which one-link stands for the legs and the other for the trunk together with the hands and the head. The figure shows also the foot, but it is considered to be immobile with respect to the support and not involved in the motion.

The motion of such a system is described by the following differential equation:

$$\begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{bmatrix} + \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}, \tag{13.11}$$

where

$$H_{11} = m_1 \cdot l_{C1}^2 + I_1 + m_2(l_1^2 + l_{C2}^2 + 2 \cdot l_1 \, l_{C2} \cdot \cos q_2) + I_2, \tag{13.12}$$

$$H_{22} = m_2 \, l_{C2}^2 + I_2, \tag{13.13}$$

$$H_{12} = H_{21} = m_2 \, l_1 \, l_{C2} \ \cos q_2 + m_2 \, l_{C2}^2 + I_2, \tag{13.14}$$

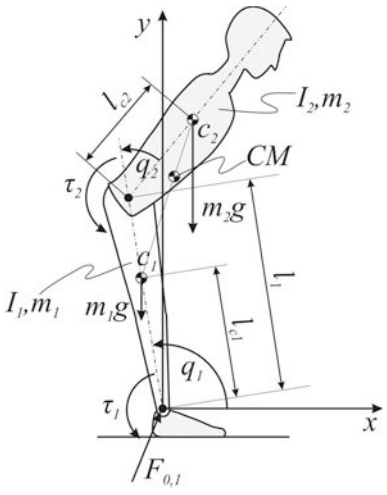


Fig. 13.9 Two-link planar mechanism

milutinn@uns.ac.rs

$$h_1 = -m_2\, l_1\, l_{C2} \cdot \sin q_2 \left((\dot{q}_2)^2 + 2\dot{q}_1\dot{q}_2 \right) + m_1\, l_{C1}\, g\ \cos q_1 + m_2\, g\ (l_{C2}\ \cos\ (q_1 + q_2) + l_1\ \cos q_1), \tag{13.15}$$

$$h_2 = m_2\, l_1\, l_{C2} \cdot \sin q_2 (\dot{q}_1)^2 + m_2\, l_{C2}\, g \cdot \cos\ (q_1 + q_2) \tag{13.16}$$

with the designation being given in Fig. 13.9. It should be noticed that H_{11}, H_{12} and H_{22} in the case considered are greater than zero. The motion of the CM of the overall system can be described as:

$$\ddot{\vec{r}}_{\mathrm{CM}} = \frac{1}{m_1 + m_2} \left(\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{bmatrix} + \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \right), \tag{13.17}$$

where

$$A_{11} = -m_1 \cdot l_{C1} \sin q_1 - m_2 l_1 \sin q_1 - m_2 l_{C2} \sin(q_1 + q_2), \tag{13.18}$$

$$A_{12} = -m_2 l_{C2} \sin(q_1 + q_2), \tag{13.19}$$

$$A_{21} = m_1 \cdot l_{C1} \cos q_1 + m_2 l_1 \cos q_1 + m_2 l_{C2} \cos(q_1 + q_2), \tag{13.20}$$

$$A_{22} = m_2 l_{C2} \cos(q_1 + q_2), \tag{13.21}$$

$$a_1 = -m_1 \cdot l_{C1} \dot{q}_1^{\,2} \cos q_1 - m_2 l_1 \dot{q}_1^{\,2} \cos q_1 - m_2 l_{C2} (\dot{q}_1 + \dot{q}_2)^2 \cos(q_1 + q_2), \tag{13.22}$$

$$a_2 = -m_1 \cdot l_{C1} \dot{q}_1^{\,2} \sin q_1 - m_2 l_1 \dot{q}_1^{\,2} \sin q_1 - m_2 l_{C2} (\dot{q}_1 + \dot{q}_2)^2 \sin(q_1 + q_2). \tag{13.23}$$

Since we deal with a planar model, the ZMP position can be calculated in the following way

$$\mathrm{ZMP}_X = \tau_1 / F_{0,1Y}, \tag{13.24}$$

where $F_{0,1Y}$ denotes the vertical component of the ground reaction force, which can be calculated as:

$$F_{0,1Y} = (m_1 + m_2)\, \ddot{r}_{\mathrm{CMY}} + (m_1 + m_2) g = [A_{21} \quad A_{22}] \begin{bmatrix} \ddot{q}_1 \\ \ddot{q}_2 \end{bmatrix} + a_2 + (m_1 + m_2) g. \tag{13.25}$$

13.5.3 Compensation Involving Only the Ankle Joint

When the compensation is performed only by the torque at the ankle joint, the hip joint remains stiff, and the values of joint-space coordinates, velocities and

milutinn@uns.ac.rs

accelerations are zero ($q_2 = 0$, $\dot{q}_2 = 0$, $\ddot{q}_2 = 0$). In that case, the system moves as an inverted physical pendulum and Eq. (13.11) reduces to:

$$\ddot{q}_1 = \frac{\tau_1 - h_1}{H_{11}}, \tag{13.26}$$

The relationship between the ZMP position and driving torque in the case when the hip joint is stiff is given by the following relation:

$$\tau_1 = \mathrm{ZMP}_X(L_{\mathrm{CM}}\,\ddot{q}_1 \cos q_1 - L_{\mathrm{CM}}\dot{q}_1^{\,2} \sin q_1 + (m_1 + m_2)g), \tag{13.27}$$

where

$$L_{\mathrm{CM}} = m_1 \cdot l_{C1} + m_2 l_1 + m_2 l_{C2}. \tag{13.28}$$

By combining Eqs. (13.15), (13.26), and (13.27) one obtains the relationship between the angular acceleration at the ankle and ZMP position as:

$$\begin{aligned}(H_{11} - \mathrm{ZMP}_X L_{\mathrm{CM}} \cos q_1)\,\ddot{q}_1 \;&= -\mathrm{ZMP}_X L_{\mathrm{CM}} \dot{q}_1^{\,2} \sin q_1 \\ &+ \mathrm{ZMP}_X (m_1 + m_2)g - g\; L_{\mathrm{CM}} \cos q_1.\end{aligned} \tag{13.29}$$

Let us assume that in the initial moment the humanoid stands still and that this state is suddenly perturbed by a force as shown in Fig. 13.8a. Due to the momentum thus received, the humanoid will start to rotate around the ankle joint at an angular velocity[1] $\dot{q}_1$ in the direction indicated in Fig. 13.10 (this direction is designated as the negative one) and the CM will move in the direction of the action of the disturbance force (positive direction of the x-axis), causing the CM displacement to the toes, i.e., to the edge of the support area. In order to recover from the disturbance, the system has to generate a negative acceleration of the CM in the direction of the x-axis, which is achieved by a positive angular acceleration at the ankle. Based on (13.26), it can be seen that the increase of the torque at the ankle generates a positive acceleration that will stop further inclination of the humanoid and prevent the CM projection from going outside the support area. However, as is evident from (13.27), the torque at the ankle influences in a direct way the position of the ZMP, so that the increase in the torque yields the increase in the x-coordinate of the ZMP position, which means that the ZMP will shift toward the toes, as shown in Fig. 13.10. To preserve dynamic balance it is necessary to keep the ZMP within the support area (must not reach its edge), so that there will exist a maximum torque that can be applied at the ankle. If the torque applied is of higher intensity, the system will rotate around the toes and dynamic balance will be lost. On the other hand, if the torque is insufficient to prevent the CM motion, this can again lead to the fall, or some other strategy has to be applied, e.g., stepping.

[1] Its intensity depends on the momentum applied to the body and humanoid's mass.

milutinn@uns.ac.rs

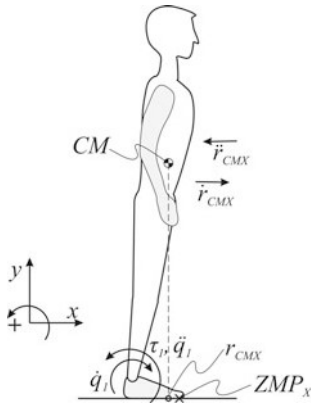


Fig. 13.10 System motion during the compensation by the ankle

To summarize, in order to preserve dynamic balance, the ZMP must remain within the support area, and the ZMP position is influenced by all the forces (gravitational and motion-induced) acting on the system. To preserve dynamic balance during the compensating movement (while the motion persists), the ZMP must be within the support area, but the CM projection need not to. At the end of the movement (when the motion stopped), the ground reaction is equal to the gravitational force, which can be seen from Eq. (13.25). Based on that and Eq. (13.24), it can be concluded that the ZMP position depends solely on gravitational forces. In that case, the ZMP and CM projection (CMP) coincide, so that the CMP will also be within the support area.

From this it follows that the driving torque at the ankle must not be greater than the value determined by the admissible ZMP displacement, so that the intensity of acceleration/deceleration of the system's CM will have its maximum allowed value. If the maximum permissible torque is not sufficient to stop the motion of the CM before its projection comes outside the support area, such strategy is not appropriate. In such situation, it is necessary to employ some other way to prevent the fall (to use different compensation strategy), for example, by the simultaneous motion at the ankle and at the hip. When the CMP comes outside the support area, then it holds that:

$$\mathrm{ZMP}_X < (m_1 + m_2)^{-1} L_{\mathrm{CM}} \cos q_1. \tag{13.30}$$

By inserting this inequality into expression (13.29) it can be seen that the angular acceleration at the ankle is negative, which causes additional acceleration of the CM. And this is just the reason why it is not possible to stop the motion of the CM if its projection is outside the support area. It should be noticed that the choice of the compensation strategy must be made immediately in the beginning of the disturbance action. Man is to learn how to do this, and it is believed that the

milutinn@uns.ac.rs

decision is made on the basis of the simultaneous processing of more information on the intensity of the disturbance force, instantaneous posture, the state of the system, etc.

13.5.4 Compensation Simultaneous Motion of the Ankle and the Hip Joints

Let us have a look again at the humanoid standing still upright, with no disturbance force acting on it (Fig. 13.8a). The action of the disturbance will transfer to it a certain momentum, and the CMP will move in the direction of the disturbance (positive direction of the x-axis).

Let us suppose then that the humanoid performs the compensating movements simultaneously at the ankle and at the hip as is shown in Fig. 13.11. The motion at the ankle joint will directly contribute to the displacement of the system's CM in the direction that is opposite to the motion caused by the disturbance, which brings about the settling of the system.

In this case both joints of the model shown in Fig. 13.9 are active. Based on the equation of the system's motion (13.11) we get:

$$\ddot{q}_1 = \frac{1}{H_{11}H_{22} - H_{21}H_{12}}\left(H_{22}(\tau_1 - h_1) - H_{21}(\tau_2 - h_2)\right), \tag{13.31}$$

$$\ddot{q}_2 = \frac{1}{H_{11}H_{22} - H_{21}H_{12}}\left(-H_{12}(\tau_1 - h_1) + H_{11}(\tau_2 - h_2)\right). \tag{13.32}$$

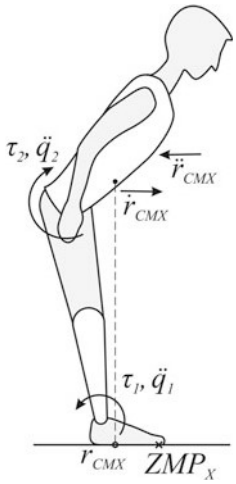


Fig. 13.11 Compensation performed by the ankle and hip joints

milutinn@uns.ac.rs

For a two-link planar mechanism it holds that $H_{11}H_{22} - H_{21}H_{12} > 0$, $H_{ii} > 0$, so that on the basis of (13.31) it can be concluded that if the torque at the ankle (τ_1) is greater than h_1 and the torque at the hip (τ_2) is smaller than h_2, the angular acceleration at the ankle will be positive. This means that this acceleration will tend to rotate the system backward. Similarly, based on Eq. (13.32), we conclude that if the torque at the ankle is greater than h_1, and that at the hip smaller than h_2, the angular acceleration at the hip will be negative. This means that in this case the trunk will rotate forward around the hip joint. It should be noticed that h_1 and h_2 are the moments generated by the Coriolis, centrifugal and gravitational accelerations at the ankle and hip joints. When the angular velocities are small, then, based on Eqs. (13.15) and (13.16), we can take that h_1 and h_2 are approximately equal to the moments generated by the gravitational forces at the ankle and hip, respectively. In order to stop the motion, the acceleration of the humanoid's CM in the x-direction must be negative, as shown in Fig. 13.11. Since A_{11} is by absolute value significantly greater than A_{12} (Eqs. 13.18 and 13.19), the effect of the ankle motion on the CM motion is much greater than the effect on the motion of the trunk (Eq. 13.17). By the same token, in order to stop the system it is necessary to generate a positive angular acceleration at the ankle. Also, the ZMP position is directly related to the driving torque at the ankle, which is given by the following equation:

$$\begin{aligned}\tau_1 = \ & \mathrm{ZMP}_X((m_1 C1 \cos q_1 + m_2 l_1 \cos q_1 + m_2 l_{C2} \cos(q_1 + q_2))\,\ddot{q}_1 \\ & + m_2 l_{C2} \cos(q_1 + q_2)\,\ddot{q}_2 \ - m_1 \cdot l_{C1} \dot{q}_1^{\,2} \sin q_1 - m_2 l_1 \dot{q}_1^{\,2} \sin q_1 \\ & - m_2 l_{C2} (\dot{q}_1 + \dot{q}_2)^2 \sin(q_1 + q_2) + (m_1 + m_2) g).\end{aligned} \tag{13.33}$$

As in the previous case, this equation provides a limit to the maximum τ_1 that may be applied at the ankle. However, the maximum torque τ_1 does not guarantee that the angular acceleration at the ankle will be positive and that CM will decelerate. The maximum angular acceleration of the CM for the case when only the ankle joint is active is:

$$\ddot{q}_1^{\mathrm{MAX}} = \frac{\tau_1^{\mathrm{MAX}} - h_1}{H_{11}}. \tag{13.34}$$

Maximum angular acceleration when both the ankle and the hip are active is:

$$\ddot{q}_1^{\mathrm{MAX}} = \frac{1}{H_{11}H_{22} - H_{21}H_{12}} \left(H_{22}(\tau_1^{\mathrm{MAX}} - h_1) - H_{21}(\tau_2^{\mathrm{MIN}} - h_2) \right). \tag{13.35}$$

It is easy to show that:

$$\frac{H_{22}}{H_{11}H_{22} - H_{21}H_{12}} > \frac{1}{H_{11}} \tag{13.36}$$

milutinn@uns.ac.rs

from which it follows that the same torque at the ankle joint in the case when both joints are active can generate a much greater angular acceleration at the ankle than when only this joint is active. Therefore, the simultaneous motion of the ankle and the hip may produce a significantly greater deceleration of the CM motion that is to compensate for a significantly larger disturbance. Besides, this way allows simultaneous deceleration of the CM motion and control of the ZMP position, so that if during the compensating movement the CM projection is found outside the support area, simultaneous motion at the ankle and the hip can "pull in" the CM projection into the support area. When the compensation is performed by the motion only at the ankle, there will be a direct relationship between the angular acceleration at the ankle (and thus the CM acceleration) and ZMP position (Eq. 13.29). Since the ZMP must be within the support area, this narrows the margin for the deceleration of the CM motion in the x-direction. And these are just the reasons why the compensation involving both the ankle and the hip can compensate for the disturbances of greater intensity compared to the case when compensation is performed by the ankle only.

However, this strategy has also its limitations. From (13.25) one can see that the angular accelerations at the ankle and hip are limited because the foot-ground contact is unilateral (the foot can push, but not pull, the support), so that the vertical component of the ground reaction force is always directed upwards. Another constraint comes out from the maximum amplitude of the torque that can be generated by the actuators at the hip.

13.5.5 Compensation Simulation—Only by Ankle and Using Simultaneously Ankle and the Hip Joints

In the first case we consider the realization of compensation using only the ankle. In the initial moment, the robot stands still with both feet on the ground. The torques at the joints (there exists only gravitational loads) are constant, so that the control inputs are also constant and we can write:

$$u_j^{t_i} = u_j^0, \quad j \in \{2, ..., 46\}, \tag{13.37}$$

where $u_j^{t_i}$ is the motor voltage activating the joint j at the moment t_i, whereas u_j^0 is the reference control voltage, which in this example corresponds to the gravitational load of the joints prior to the disturbance. At a certain moment, the disturbance force of an amplitude of 510 N and profile shown in Fig. 13.8a, starts to act in the direction of the x-axis. The force transfers a momentum of 18 kgm/s, which brings about the CM motion at a speed of 0.21 m/s.

Immediately after the occurrence of the disturbance, the compensation by the action at the ankle is to start, to stop the CM motion in the x-direction, but so that dynamic balance is preserved (the ZMP has to remain within the support area). The

milutinn@uns.ac.rs

deviation of the CM from its initial position, desired ZMP position,[2] and the ZMP position error are calculated in the following way:

$$\Delta \mathrm{CM}_X^{t_i} = \mathrm{CM}_X^0 - \mathrm{CM}_X^{t_i}, \tag{13.38}$$

$$\begin{aligned} {}^{\mathrm{ref}}\mathrm{ZMP}_X^{t_i} = {}& {}^{\mathrm{ref}}\mathrm{ZMP}_X^{t_i-1} + \Delta \mathrm{CM}_X^{t_i}\left(K_{\mathrm{pcz}} + K_{\mathrm{icz}} + K_{\mathrm{dcz}}\right) \\ & - \Delta \mathrm{CM}_X^{t_i-1}\left(K_{\mathrm{pcz}} - 2K_{\mathrm{dcz}}\right) + \Delta \mathrm{CM}_X^{t_i-2} K_{\mathrm{dcz}}, \end{aligned} \tag{13.39}$$

$$\Delta \mathrm{ZMP}_X^{t_i} = {}^{\mathrm{ref}}\mathrm{ZMP}_X^{t_i} - \mathrm{ZMP}_X^{t_i}. \tag{13.40}$$

K_{pcz} represents the proportional feedback gain aiming to control the CM position via the ZMP position, while K_{dcz} and K_{icz} are the derivative and integral gains used in (13.39). In view of the fact that the system is symmetric at both ankles (joints 8 and 16 in Fig. 13.3), the same control voltage is applied, calculated in the following way:

$$\begin{aligned} \Delta u_{8,16}^{t_i} = {}& \Delta u_{8,16}^{t_i-1} + \Delta \mathrm{ZMP}_X^{t_i}\left(K_{\mathrm{pz}} + K_{\mathrm{dz}} + K_{\mathrm{iz}}\right) \\ & - \Delta \mathrm{ZMP}_X^{t_i-1}\left(K_{\mathrm{pz}} + 2K_{\mathrm{dz}}\right) + \Delta \mathrm{ZMP}_X^{t_i-2} K_{\mathrm{dz}}, \end{aligned} \tag{13.41}$$

$$u_{8,16}^{t_i} = u_{8,16}^{t_i-1} + \Delta u_{8,16}^{t_i}, \tag{13.42}$$

where $\Delta u_{8,16}^{t_i}$ designates the control (voltage) increment at the moment t_i (joints 8 and 16) calculated in the form of a PID regulator minimizing the deviation of the ZMP position. Such obtained control increment is added to the control from the preceding time instant. Expression (13.41) is obtained in a similar way as (13.39). K_{pz}, K_{dz} and K_{iz} represents proportional, derivative and integral gains with respect to the deviation of ZMP position, used in (13.41).

All the joints apart from two ankles should be stiff. Because of that, the actuators at these joints are velocity-controlled, and the desired velocity is zero. The control law is:

[2]The ZMP position when the compensating action is ended coincides with the CM position. Hence, it is important that the ZMP position to which the system returns (denoted here as the reference one) is defined so as to minimize the deviation of the actual CM position from its position prior to the action of the disturbance. This can be done in the form of the PID regulator in the following way:

$${}^{\mathrm{ref}}\mathrm{ZMP}_X^{t_i} = K_{\mathrm{pcz}} \Delta \mathrm{CM}_X^{t_i} + K_{\mathrm{icz}} \sum_{j=1}^{i} \Delta \mathrm{CM}_X^{t_j} + K_{\mathrm{dcz}}\left(\Delta \mathrm{CM}_X^{t_i} - \Delta \mathrm{CM}_X^{t_{i-1}}\right)$$

Expression (13.39) is obtained by subtracting the above expression, written for the time instant t_{i-1}, from the expression written for the time instant t_i.

milutinn@uns.ac.rs

$$\Delta\dot{q}_j^{t_i} = -\dot{q}_j^{t_i}, \tag{13.43}$$

$$\begin{aligned} u_j^{t_i} = \; & u_j^{t_i-1} + \Delta\dot{q}_j^{t_i}\left(K_{\mathrm{pvj}} + K_{\mathrm{ivj}} + K_{\mathrm{dvj}}\right) - \Delta\dot{q}_j^{t_i-1}\left(K_{\mathrm{pvj}} - 2K_{\mathrm{dvj}}\right) \\ & + \Delta\dot{q}_j^{t_i-2}K_{\mathrm{dvj}}, \quad j \in \{2..46\}\backslash\{8, 16\}. \end{aligned} \tag{13.44}$$

K_{pvj} represents proportional, K_{dvj} derivative and K_{ivj} integral gain to control the velocity at the joint j. It should be noted that these gains are different for each of the joints (Table 13.3).

Figure 13.12a shows the resulting humanoid motion obtained by the simulation. The CM position prior to the disturbance coincides with the reference one. Since only the ankle joint is active, the angle at this joint prior to the disturbance is equal to the angle at the instant when the CM position attains its reference value. From this, it comes out that the initial posture is identical to the final one, as shown in Fig. 13.12a. The figure shows also the posture when the robot is inclined most, and this corresponds to the moment when the CM velocity in the x-direction drops to

Table 13.3 Gains used in the simulation

	CM position feedback gains (Eq. 13.39)				ZMP position feedback gains (Eq. 13.41) applied at ankle joints		
	K_{pcz}	K_{dcz}	K_{icz}	Joint	K_{pz}	K_{dz}	K_{iz}
	15.64	7750	0.0022	8 and 16	15.64	39.72	0.4443
				Gains for the regulation of the angular velocity at the joint			
Joint	K_{pvj}	K_{dvj}	K_{ivj}	Joint	K_{pvj}	K_{dvj}	K_{ivj}
2 and 10	112.3	142.7	6.38	27	541.7	688	30.8
3 and 11	178.3	170.1	13.48	28	545.2	692.3	31.0
4 and 12	140.2	133.6	10.61	29	543.3	689.9	30.9
5 and 13	174.8	222	9.93	30	544.3	691.2	30.9
6 and 14	60.89	77.35	3.45	31	544.9	692.0	31.0
7 and 15	63.5	80.67	3.607	32	546.9	694.6	31.1
8 and 16	64.1	81.42	3.64	33	546.6	694.2	31.1
9 and 17	59.34	75.38	3.37	34	532.4	676.1	30.2
18	144.4	137.6	10.9	35	534.9	679.3	30.4
19	180.3	171.8	13.7	36	599.2	761.1	34.0
20	597.6	758.9	33.9	37	604.3	767.6	34.3
21	592.6	752.7	33.7	38	82.0	104.2	4.7
22	533.2	677.2	30.3	39 and 43	217.8	276.6	12.4
23	535	679.6	30.4	40 and 44	62.4	79.2	3.5
24	547.2	695	31.1	41 and 45	91.6	116.4	5.2
25	547.7	695.7	31.1	42 and 46	69.6	88.5	4.0
26	544.4	691.4	30.9				

milutinn@uns.ac.rs

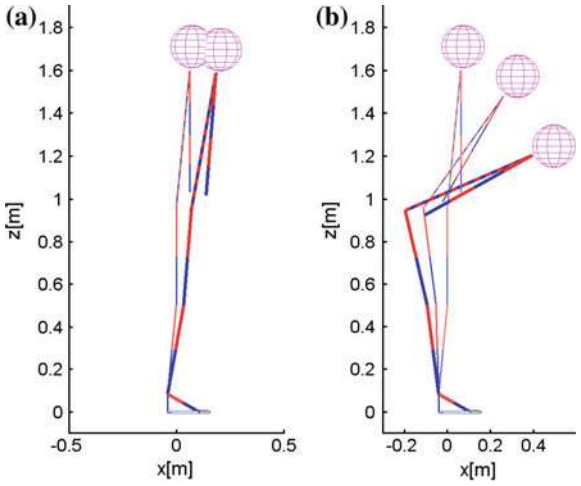


Fig. 13.12 **a** Compensating movement obtained by the ankle joint, **b** by the ankle and hip joints simultaneously

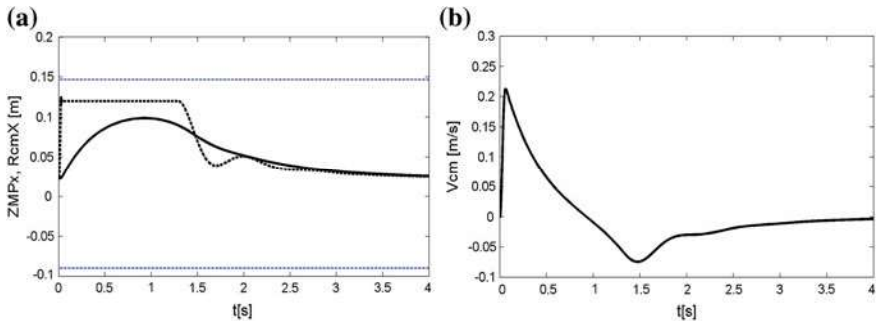


Fig. 13.13 **a** ZMP position (*thick dashed line*) and CM position (*thick full line*), foot borders (*thin dashed line*) and **b** CM velocity

zero, after which the robot gradually returns to the initial posture. Figure 13.13a shows the motion of the ZMP and CM in the x-direction, whereas Fig. 13.13b shows the velocity of the CM motion in the same direction.

As can be seen from Fig. 13.13b, there is at the beginning an abrupt jump of the CM velocity, which is a consequence of the disturbance force influence on the trunk. The maximum reference ZMP position ($^{\text{ref}}\text{ZMP}_X^{\text{MAX}}$) is selected so that its distance to the foot front edge is 25 mm. This ensures that in the case of an overshoot or disturbance, control system can keep the ZMP within "safe area" and the system remains dynamically balanced. After the occurrence of the disturbance, the reference

milutinn@uns.ac.rs

ZMP position becomes equal to maximum allowed one. The actual ZMP attains quickly the reference position (Fig. 13.13a). After approximately 1 s the CM velocity drops to zero (Fig. 13.13b), after which the CM returns gradually to its initial position, and after about 4 s the robot settles down, and the CM position coincides with the position that existed prior to the action of the disturbance. If the disturbance force increases, and the CM moves in the x-direction, further from the maximum allowed ZMP position, the angular acceleration will become positive and the robot will fall down. As is evident from Fig. 13.13a, the CM position is close to the ZMP position, which means that the intensity of the applied disturbance force was close to the maximum that can be compensated for by applying this strategy. In the second case, the disturbance compensation is performed by simultaneous movement at the ankle and hip joints. In the initial moment, the robot stands still with both feet on the ground. Control applied before disturbance start is the same as in the previous case (Eq. 13.37).

The second case, we consider is compensation by ankle and hip joints simultaneously. The disturbance force acting on the robot produces a momentum of 43 kgm/s, which will cause the movement of the CM of the overall system. The disturbance is immediately followed by the compensating movement, simultaneously at both the ankle and the hip. Thus, with the aid of the ankle it will be possible again to regulate the ZMP position (Eqs. 13.41 and 13.42), but in this case we will adopt the reference ZMP position to be the maximum one that is allowed by the foot dimension:

$$^{\mathrm{ref}}\mathrm{ZMP}_X^{t_i} = {}^{\mathrm{ref}}\mathrm{ZMP}_X^{\mathrm{MAX}}. \tag{13.45}$$

Instead of controlling the motion of the hip (joints 4 and 12), the inclination will be performed by the first joint of trunk (joint 18). By the trunk motion is regulated the position of the CM projection, and itis reference position is at the same place as prior to the occurrence of the disturbance. The control for this joint can be written in the form:

$$u_{18}^{t_i} = u_{18}^{t_i-1} + \Delta\mathrm{CM}_X^{t_i}\left(K_{\mathrm{pc}} + K_{\mathrm{ic}} + K_{\mathrm{dc}}\right) - \Delta\mathrm{CM}_X^{t_i-1}\left(K_{pc} - 2\mathrm{Kdc}\right) + \Delta\mathrm{CM}_X^{t_i-2}K_{\mathrm{dc}}. \tag{13.46}$$

In the case considered, the $K_{\mathrm{pc}} = 3636$ is proportional, $K_{\mathrm{ic}} = 1.652$ integral and $K_{\mathrm{dc}} = 5.77 \cdot 10^5$ derivative gaint.

As in the previous case, all the joints apart from the two ankle joints (8 and 16) and the first trunk joint (18), are immobile. The motion of the ankle joints is controlled according to Eqs. (13.41) and (13.42), the control of the trunk joint (18) is given by Eq. (13.46), whereas all remaining joints are stiff and their control is performed according to Eqs. (13.43) and (13.44).

When the velocity of the CM becomes sufficiently small, the compensation is (say at the instant T) switched to the action of the ankle joint only. The motion of the first link of the trunk (joint 18) should be stopped, and this has to be done

milutinn@uns.ac.rs

gradually, to avoid abrupt acceleration change and undesired effects on the ZMP position. Hence the control of this joint will be velocity-based:

$$\Delta\dot{q}_{18}^{t_i} = \left(1 - f\left(\frac{t_i - T}{0.8s}\right)\right)\dot{q}_{18}^{T} - \dot{q}_{18}^{t_i}, \tag{13.47}$$

$$u_{18}^{t_i} = u_{18}^{t_i-1} + \Delta\dot{q}_{18}^{t_i}\left(K_{\mathrm{pv18}} + K_{\mathrm{iv18}} + K_{\mathrm{dv18}}\right) - \Delta\dot{q}_{18}^{t_i-1}\left(K_{\mathrm{pv18}} - 2K_{\mathrm{dv18}}\right) + \Delta\dot{q}_{18}^{t_i-2}K_{\mathrm{dv18}}. \tag{13.48}$$

The gains for the regulation of angular velocity at the joint 18 are given in Table 13.3. In this case, the reference velocity changes with time. The function f ensures a smooth convergence of the instantaneous velocity to zero. Since this function in the initial moment is zero, the reference velocity is equal to the instantaneous one. After 0.8 s, (predefined time for the trunk deceleration), the value of the function f becomes 1, so that the reference velocity at the joint 18 will be reduced to zero. Of course, to drive the joint 18 to such state appropriate control (Eq. 13.48) is needed.

Control by the ankle and calculation of the reference ZMP position are the same as in the case of compensation by the ankle only (Eqs. 13.38–13.42). All the joints apart from the two ankles joints (8 and 16) and first joint of the trunk (18) should remain stiff, and the control law for these joints will not change (Eqs. 13.43 and 13.44).

Figure 13.12b depicts the movement obtained in three characteristic moments. It shows the posture when the disturbance force is starting, the moment when the compensation by the ankle and hip switches to the compensation by the ankle only, and the posture when the system stopped.

Figure 13.14a shows the positions of the ZMP and CM in the x-direction, whereas Fig. 13.14b shows the CM velocity in the same direction. In both figures is marked the moment when the simultaneous compensation by the ankle and hip switches to the compensation by the ankle only.

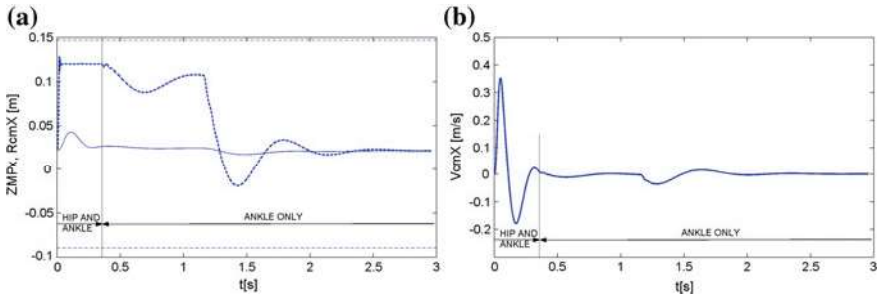


Fig. 13.14 **a** ZMP position is shown by the *thick dashed line* and CM position by the *thin full line*. *Thin dashed lines* show the foot borders **b** CM velocity. Also, it is shown the moment when the compensation by the ankle and hip switches to the compensation by the ankle only

milutinn@uns.ac.rs

Like in the previous case, an abrupt jump of the CM velocity can be seen (Fig. 13.13b) in the beginning of the disturbance action. By comparing Figs. 13.13b and 13.14b one can see that in the case of the compensation by the simultaneous movement at the ankle and the hip, the CM velocity drops much faster to zero, although the disturbance intensity was more than twice higher than in the previous case. As can be seen from Fig. 13.13a, the CMP shifts very little during the whole movement and in none of the instants there exists a danger that the CM comes out the support area. However, Fig. 13.11b shows large inclination of the trunk (almost 90°) what does not correspond to behavior of humans we may often observe. However, what is the disturbance intensity that can be compensated for this way, and when it is necessary to switch to the compensation by stepping, it is still an open issue. The answer to this question should be obtained by the future research.

13.6 Conclusions

In this paper, we showed that the two types of disturbances may act on humanoid robot: small and large. Each of them should be compensated for in a different way.

When considering small disturbances the compensation should be distributed onto the compensation of internal synergy (preservation of a desired form of motion) and compensation aimed at preserving dynamic balance. When there is no danger of falling, it is necessary to intensify the activity on compensation of internal synergy; when dynamic balance is endangered an urgent action is needed to prevent fall, while all other actions can be postponed. An indicator of the necessity of intensifying one or the other compensational action is the ZMP. The work demonstrated the application of a simple and efficient way of changing the "engagement degree" of corrective control for compensating internal synergy and preservation of dynamic balance.

In case of the compensation for large disturbances, it is of absolute priority to prevent the fall, i.e., to ensure that the ZMP position during the walk remains within the support area. Since the disturbances are of higher intensity, the compensating movements have to be appropriate, and the main role in them play an inertial forces. However, their action ends after the realization of the "energetic part" of the movement (which is always very short), and the moments of gravitational forces take over the dominant influence on the ZMP position. As a consequence, the compensating movement has to be such that at its end the humanoid remains in the position that ensures its static balance.

Acknowledgements This work was in part supported by the Ministry of Science and Technological Development of the Republic of Serbia under contract III44008, and in part by the Autonomous Province of Vojvodina, Provincial Secretariat for Science and Technological Development, Serbia, under contract 114-451-2116/2011-03.

milutinn@uns.ac.rs

References

1. Fukuda, T., Michelini, R., Potkonjak, V., Tzafestas, S., Valavanis, K., Vukobratović, M.: How far away is "artificial man". IEEE Robot. Autom. Mag. 66–73 (2001)
2. Juričić, D., Vukobratović, M.: Mathematical Modeling of Biped Walking Systems. ASME Publlications. 72-WA/BHF-13 (1972)
3. Vukobratović, M.: How to control the artificial anthropomorphic systems. IEEE Trans. Syst. Man Cybern. SMC-3, 497–507 (1973)
4. Vukobratović, M.: Legged Locomotion Systems and Anthropomorphic Mechanisms, Mihajlo Pupin Institute, Belgrade, (1975); also published in Japanese by Nikkan Shimbun Ltd. Tokyo (1976), in Russian by "MIR", Moscow (1976), in Chinese, Beijing (1983)
5. Vukobratović, M., Borovac, B., Surla, D., Stokić, D.: Biped Locomotion—Dynamics, Stability Control and Application. Springer, Berlin (1990)
6. Vukobratović, M., Borovac, B.: Zero-moment point—thirty five years of its life. Int. J. Humanoid Rob. **1**(1), 157–173 (2004)
7. Vukobratović, M., Borovac, B.: Note on the article, zero-moment point—thirty five years of its Life. Int. J. Humanoid Rob. **2**(2), 225–227 (2005)
8. Breniere, Y., Bril, B.: Development of postural control of gravity forces in children during the first 5 years of walking. Exp. Brain Res. **121**(3), 255–262 (1998)
9. Vukobratović, M., Borovac, B., Potkonjak, V.: Towards a unified understanding of basic notions and terms in humanoid robotics. Robotica **25**, 87–101 (2007)
10. Vukobratović, M., Borovac, B., Raković, M., Potkonjak, V., Milinović, M.: On some aspects of humanoid robots gait synthesis and control at small disturbances. Int. J. Humanoid Rob. **5** (1), 225–227 (2008)
11. Borovac, B., Nikolić, M., Raković, M.: How to compensate for the disturbances that jeopardize dynamic balance of a humanoid robot? Int. J. Humanoid Rob. **8**(3), 533–578 (2011)
12. Vukobratović, M., Potkonjak, V., Borovac, B., Babković, K.: Simulation model of general human and humanoid motion. Multibody Sys. Dyn. **17**(1), 71–96 (2007)

milutinn@uns.ac.rs